Assignment 10

Total = 10 Marks

Multiple Choice Questions

Question 1: A dataset contains $m = 27$ predictors. Using the commonly adopted default Random Forest settings, which one of the following is the most suitable values of p and n_{min}?

Here, p denotes the number of predictors randomly selected at each split, and n_{min} denotes the minimum number of observations in a leaf node.

(a) $p = 9,\ n_{min} = 5$ for regression problems
(b) $p = 5,\ n_{min} = 1$ for classification problems
(c) $p = 5,\ n_{min} = 5$ for regression problems
(d) $p = 9,\ n_{min} = 1$ for classification problems

[2 Mark]

Correct answer: (Multiple correct Answer)
Both
(a) $p = 9,\ n_{min} = 5$ for regression problems
and
(b) $p = 5,\ n_{min} = 1$ for classification problems
are correct.
Partial marking as follows: if only one of (a) or (b) selected, award 1 mark, if both (a) and (b) selected award 2 marks, and in all other cases award 0 marks.

Detailed Solution: For regression, the default number of predictors is

$$p = largest\ integer \leq \left(\frac{m}{3}\right) = \left(\frac{27}{3}\right) = 9$$

$$n_{min} = 5$$

For classification, the default number of predictors is

$$p = largest\ integer \leq \sqrt{\mathrm{m}} = \sqrt{27} = 5.196$$

Therefore, $p = 5$

$$n_{min} = 1$$

Question 2: A binary-classification node in a Decision Tree contains 10 observations: 5 from Class A and 5 from Class B. Two candidate splits produce as follows:

Split 1:
Left child: 4 (Class A) and 1 (Class B)
Right child: 1 (Class A) and 4 (Class B)

Split 2:

Left child: 3 (Class A) and 0 (Class B)
Right child: 2 (Class A) and 5 (Class B)

The Gini impurity of Split 1 and Split 2, respectively, are:

(a) 0.29 and 0.32
(b) 0.32 and 0.29
(c) 0.32 and 0.41
(d) 0.41 and 0.29

[2 Mark]

Correct answer: (b) 0.32 and 0.29

Detailed Solution:

For a binary node,

$$G = p_A(1 - p_A) + p_B(1 - p_B)$$

For Split 1, each child contains five observations.
Left child:

$$G_L = \frac{4}{5}\left(1 - \frac{4}{5}\right) + \frac{1}{5}\left(1 - \frac{1}{5}\right) = 0.32$$

Right child has the reversed class proportions, so

$$G_R = 0.32$$

Hence,

$$G_{Split1} = \frac{5}{10}(0.32) + \frac{5}{10}(0.32) = 0.32$$

For Split 2, the left child is pure:

$$G_L = 0$$

For the right child,

$$p_A = \frac{2}{7}, \quad p_B = \frac{5}{7}$$

Thus,

$$G_R = \frac{2}{7}\left(1 - \frac{2}{7}\right) + \frac{5}{7}\left(1 - \frac{5}{7}\right) = \frac{20}{49} \approx 0.41$$

The weighted impurity is

$$G_{Split2} = \frac{3}{10}(0) + \frac{7}{10}(0.408) \approx 0.29$$

Question 3: In a gradient boosting based regression model, suppose the current model predictions at a particular boosting stage are $f_{j-1}(x_i) = [6,8,11,13]$ and the observed responses are, $y_i = [8,7,13,12]$. Using the squared-error loss $J = \sum_i \frac{1}{2}\left(y_i - f(x_i)\right)^2$,

determine which vector represents the negative gradients to be approximated by the next weak learner?

(a) $[-4, \quad 2, \quad -4, \quad 2]$

(b) $[4, \quad -2, \quad 4, \quad -2]$

(c) $[2, \quad -1, \quad 2, \quad -1]$

(d) $[-2, \quad 1, \quad -2, \quad 1]$

[3 Mark]

Correct answer: (c) $[2, \quad -1, \quad 2, \quad -1]$

Detailed Solution:
For squared-error loss,

$$J_i = \frac{1}{2}\left(y_i - f(x_i)\right)^2$$

Differentiating with respect to the prediction gives

$$g_{ij} = \left(f_{j-1}(x_i) - y_i\right)$$

For Observation 1,

$$g_{1j} = (6 - 8) = -2$$

For Observation 2,

$$g_{2j} = (8 - 7) = 1$$

For Observation 3,

$$g_{3j} = (11 - 13) = -2$$

For Observation 4,

$$g_{4j} = (13 - 12) = 1$$

Hence,

$$g_j = [-2, 1, -2, 1]$$

Gradient boosting fits the next weak learner to the negative gradient, so

$$-g_j = [2, \quad -1, \quad 2, \quad -1]$$

Therefore, $[2, \quad -1, \quad 2, \quad -1]$ gives the negative gradients to be approximated by the next weak learner.

Question 4: Consider a CART model for regression with one input (x) and one output (y) variable. The following data is given:

For $x = [2, 4, 10, 12]$ corresponding response values of y are: $y = [4, 6, 18, 20]$

A candidate CART split is made at $x < 7$. Determine the constant predictions c_1 and c_2 in the two regions and the overall MSE after the split.

(a) $c_1 = 5, c_2 = 19, MSE = 1$
(b) $c_1 = 3, c_2 = 11, MSE = 1$
(c) $c_1 = 5, c_2 = 19, MSE = 2$
(d) $c_1 = 3, c_2 = 11, MSE = 2$

[3 Mark]

Correct answer: (a) $c_1 = 5$**,** $c_2 = 19$**,** $MSE = 1$

Detailed Solution: The split $x < 7$ divides the observations into two regions.

For Region 1, $x < 7$, the response values are $y = \{4,6\}$. Therefore, the constant CART prediction is the mean response:

$$c_1 = \frac{4 + 6}{2} = 5$$

The MSE within region 1 is, $E_1 = \frac{(4-5)^2+(6-5)^2}{2} = 1$

For region 2, $x \geq 7$, the response values are $y = \{18,20\}$. Hence,

$$c_2 = \frac{18 + 20}{2} = 19$$

The MSE within region 2 is, $E_2 = \frac{(18-19)^2+(20-19)^2}{2} = 1$

Since each region contains two observations, the overall MSE is,

$$MSE = \frac{2E_1 + 2E_2}{4} = 1$$

Therefore, $c_1 = 5$, $c_2 = 19$, $MSE = 1$.